

Assignment No. 4

III B. Tech (CSE)-Prob & Stat

1. What are the different measures of dispersion? Which one is an ideal measure of dispersion and why?
2. Define coefficient of variation. For a set of 10 observations on a variable X, following calculations were made:

$$V = (X-30)/2, \sum_{i=1}^{10} V_i = -1, \sum_{i=1}^{10} V_i^2 = 49$$

It was later noticed that an observation 40 was misread as 30. Find the correct coefficient of variation of X.

3. Define the coefficient of variation and discuss its utility in statistics.
The second moment about mean 110.4 of five observations is 31.04. If the three observations of the five observations are 104, 106 and 120, find the other two.

4. A distribution is characterized as follows:

Sl.No.	Size	Mean	S.D.
1	100	15	3
2	150	25	5
3	200	30	7
4	250	?	6

Given that the mean of the combined data is 22.5, find the coefficient of variation of the combined data.

5. Define coefficient of variation. For a set of 10 observations on a variable X, following calculations were made:

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It was later noticed that an observation 40 was misread as 30. Find the correct coefficient of variation of X.

6. Define coefficient of variation and discuss its utility in Statistics.
The mean and the S.D. of 10 values were found to be 9.5 and 2.5 respectively. Later on, an additional observation becomes available. This was 15.0 and was included in the original sample. Find the coefficient of variation of the 11 observations.

7. What are the uses of β_1 and β_2 ? Show that $\beta_2 \geq 1$.

8. Define coefficient of variation and discuss its utility in Statistics.
The mean and the S.D. of 10 values were found to be 9.5 and 2.5 respectively. Later on, an additional observation becomes available. This was 15.0 and was included in the original sample. Find the coefficient of variation of the 11 observations.

9. Show that S. D. cannot be less than mean absolute deviation with respect to mean.